

INTERACTIVE AR EXPLORATION OF EARTH'S GEOLOGICAL HISTORY

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Research Abstract

Pradeepa W D R

(IT22033482)

Supervisor: Mr. Aruna Ishara Gamage

Co – Supervisor: Mr Nushkan Nizme

B.Sc. (Hons) Degree in Information Technology Specializing in
Interactive Media (IM)

Department of Information Technology

Sri Lanka Institute of Information Technology,
Sri Lanka

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ABSTRACT

Traditional museum displays often struggle to communicate the vast timescales and dynamic processes of Earth's geological history, leaving visitors with fragmented or abstract impressions. To address this challenge, a mobile Augmented Reality (AR) system is proposed that overlays interactive storytelling and 3D reconstructions onto existing exhibits in the geology gallery of the Colombo National Museum. Visitors can scan QR codes associated with the Paleozoic, Mesozoic, and Cenozoic eras to unlock immersive AR scenes that illustrate ancient environments, evolutionary milestones, and major geological transformations. Narrated visualizations explain key concepts such as mass extinctions, plate movements, and the rise of dominant species, while gamified interactions such as coloring and animating prehistoric life promote engagement, especially among younger audiences. The learning impact is assessed through pre- and post-experience quizzes that measure knowledge gain, retention, and visitor interaction levels. By combining AR visualization, narrative design, and interactive activities, the project transforms static fossil and rock displays into participatory experiences that foster curiosity, creativity, and deeper understanding of Earth's natural history.

Keywords: Augmented Reality, Geological Eras, Museum Learning, 3D Visualization, Interactive Education

Research Objectives

1. Design a mobile AR learning flow for exploring Earth's geological history in museum contexts.
2. Integrate QR/image marker access, era-based AR storytelling, AI narration, and Color-to-AR interaction.
3. Evaluate learning impact through pre- and post-experience quizzes and usability feedback.
4. Propose a low-cost, smartphone-based solution suitable for museums and educational exhibitions.

Key Results (Summary)

Metric	Result
Participants	27 (5 kids, 9 teens, 13 adults)
Pre-quiz average	58.02%
Post-learning average	62.98%
Overall improvement	+4.95 percentage points
Participants who improved	16 of 27 (59.3%)
Largest gain by age group	Children (+8.00 pp)

Conclusion

This research demonstrates a practical, affordable AR approach to making geological history more accessible, interactive, and memorable for museum visitors. The combination of visual storytelling, creative Color-to-AR activity, and quiz-based evaluation shows promise for enhancing informal science learning, particularly among younger audiences.